

Macro Seasons

Data sources, season rules and monthly portfolio construction.

At each month-end, the model calculates one Growth score, one Inflation score and one Liquidity score. Growth and Inflation set the four season weights. Liquidity changes the final mix between risk and defensive ETFs.

1 The four seasons

Season	Growth and inflation	Economic reading	Fixed allocation emphasis
SPRING	Growth rising Inflation cooling	Recovery with disinflation	Broad equities, technology, credit, housing and modest duration
SUMMER	Growth rising Inflation heating	Reflationary expansion	Cyclicals, value, energy, materials, emerging markets, commodities and TIPS
FALL	Growth falling Inflation heating	Stagflation	Gold, broad commodities, inflation protection, dollar, defensive equities and short duration
WINTER	Growth falling Inflation cooling	Disinflationary slowdown	Treasuries, defensive and low-volatility equities, gold, safe-haven currencies and cash

2 Data used for each pillar

GROWTH Data and source ALFRED: Industrial Production (INDPRO) and Nonfarm Payrolls (PAYEMS). FRED: Initial Jobless Claims (ICSA). Yahoo Finance: XLI, XLP, SPY and IEF adjusted prices. Changes used: industrial production over 6m; payrolls over 3m; claims over 3m (falling claims count as stronger growth); XLI versus XLP and SPY versus IEF over 6m.	INFLATION Data and source ALFRED: Consumer Price Index (CPIAUCSL). FRED: Five-Year Breakeven Inflation (T5YIE) and WTI crude oil (DCOILWTICO). Yahoo Finance: TIP and IEF adjusted prices. Changes used: change in 12-month CPI inflation over 3m; breakeven inflation over 3m; WTI and TIP versus IEF over 6m.	LIQUIDITY Data and source ALFRED: M2 money stock (M2SL). FRED: NFCL, high-yield OAS, 2-year Treasury yield, Fed balance sheet, reverse repos and the Treasury General Account. Changes used: NFCL level and 3m change; OAS 3m; M2 6m; 2-year yield 12m; Fed net liquidity 3m. Portfolio effect: easier liquidity can move up to 15 points from defensive ETFs to risk ETFs; tighter liquidity can move up to 25 points the other way.
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3 From month-end data to next month's portfolio

1	Select the month-end data	Use ALFRED decision-date values for CPI, industrial production, payrolls and M2. Apply the assigned publication delay to the other FRED series, which use latest-revised history. Use Yahoo adjusted prices through month-end.
2	Calculate three scores	Calculate the listed trailing changes and set their signs so positive means stronger growth, higher inflation or easier liquidity. Standardize each reading against up to 120 prior months, with at least 36 months required and values capped at three standard deviations. Average the available readings and apply a two-month exponential average.
3	Calculate four season weights	Stronger growth raises Spring and Summer; weaker growth raises Fall and Winter. Higher inflation raises Summer and Fall; lower inflation raises Spring and Winter. The four values total 100% and determine how much of each season allocation enters the portfolio. They classify current conditions rather than forecasting event probabilities.
4	Build next month's holdings	Average each fixed season allocation with a version adjusted for trailing 36-month volatility. Blend the four results using the current season percentages, then apply the liquidity, real-yield, credit, momentum, trend and portfolio-volatility rules.
5	Record next month's return	Apply the month-end weights to the following month's returns. The long-only backtest charges 10 basis points per unit of turnover. The later physical L/S simulation applies its separate trading and financing assumptions.

Reading the dashboard
The four displayed values come from the Growth and Inflation scores and sum to 100%. The portfolio uses them directly to blend the four season allocations. A 31% Summer value assigns 31% of the pre-overlay blend to the Summer allocation; the other season allocations receive the remaining 69%. The values describe the current classification and are not calibrated forecasts of future events.

Portfolio construction and validation

How the long-only and L/S portfolios are built, tested and constrained.

4 Season allocations

How the ETFs and starting weights were set

ETF choice follows each asset's economic role. The exact percentages are fixed portfolio-design choices rather than model estimates. The main expression of a season carries 12.5%-20%, most supporting exposures carry 10%, and narrower diversifiers carry 5%-7.5%.

Spring

SPY supplies broad equity exposure; QQQ and SMH add growth and technology; IWM and XHB add domestic cyclical and housing exposure; HYG and LQD add credit; IEF adds duration; GLD diversifies the allocation.

Starting weights: SPY 20%, QQQ 15%, SMH 10%, IWM 10%, XHB 7.5%, HYG 15%, LQD 7.5%, IEF 10%, GLD 5%

Fall

GLD, DBC, XLE and TIP target inflation and real assets; UUP adds dollar exposure; XLP, XLU and XLV add defensive equities; SHY and BIL keep duration short.

Starting weights: GLD 17.5%, DBC 10%, XLE 7.5%, TIP 15%, UUP 10%, XLP 10%, XLU 7.5%, XLV 7.5%, SHY 7.5%, BIL 7.5%

Summer

XLE, XLB, XLI, VLU, IWM and EEM target a reflationary expansion. DBC, CPER and GLD add commodity exposure, while TIP adds inflation-linked bonds.

Starting weights: XLE 15%, XLB 10%, XLI 10%, VLU 10%, IWM 10%, EEM 10%, DBC 12.5%, CPER 5%, GLD 7.5%, TIP 10%

Winter

TLT, IEF and SHY add Treasury duration; USMV, XLP and XLV favor defensive equities; GLD, FXY and FXF add safe-haven exposure; BIL holds cash.

Starting weights: TLT 17.5%, IEF 17.5%, SHY 7.5%, GLD 12.5%, USMV 10%, XLP 10%, XLV 10%, FXY 5%, FXF 5%, BIL 5%

5 Portfolio construction

LONG-ONLY PORTFOLIO

Weights sum to 100%. The portfolio holds long ETF positions and BIL. It has no shorts or borrowing.

1. Season allocation. Start with the fixed percentages in section 4. Divide each ETF's starting weight by its trailing 36-month volatility, rescale the result to 100%, and average it equally with the fixed allocation.
2. Season and liquidity blend. Combine all four adjusted season allocations using the Growth/Inflation season weights. The Liquidity score can move up to 15 percentage points from defensive ETFs to risk ETFs, or up to 25 points from risk to defense.
3. Real yields and credit. The six-month change in the 10-year Treasury yield less five-year breakeven inflation can move up to half of the available allocation between GLD and TLT/IEF. Credit stress turns on when high-yield OAS, or the earlier-history Baa yield spread proxy, exceeds 110% of its trailing 36-month median and has widened over three months; the model then halves the risk-asset allocation.
4. Momentum and trend. Rank current non-BIL holdings by return from 12 months ago to one month ago; the rank multiplier ranges from 0.75 to 1.25. Move each ETF below its 200-day moving average to BIL, using the 10-month average only when daily history is insufficient.
5. Portfolio volatility and cost. Estimate the current weight vector's volatility from the trailing 24 monthly returns. Scale can range from 0.50 to 1.00 around the 10% target; unused weight goes to BIL. Charge 10 basis points per unit of turnover.

This is the funded portfolio shown in the allocation table.

L/S PORTFOLIO

The portfolio combines two macro-season streams with a 13-ETF trend stream. Duplicate and opposing positions are netted before trading and financing costs are applied.

1. Two season streams. The core stream uses the season blend and Liquidity overlay. The enhanced stream uses the complete long-only process. Each stream is separately scaled between 0.50 and 1.50 using volatility through the previous month.
2. Time-series momentum stream. The 13 ETFs are SPY, QQQ, IWM, EFA, EEM, TLT, IEF, GLD, DBC, UUP, FXY, FXF and HYG. Each is long when its trailing 12-month return exceeds BIL and short otherwise.
3. Momentum sizing. Use trailing 36-month volatility so a 10%-volatility ETF receives roughly one-thirteenth of the stream. Cap each position at 20% of the stream, then scale the stream between 0.50 and 1.50 toward 10% volatility.
4. Combine streams. Weight the three streams inversely to their trailing 24-month volatility, measured through the previous month. Scale the combined return stream between 0.50 and 1.50 toward 10% volatility.
5. Execution and costs. Add duplicate positions, cancel opposing exposures, reduce BIL before recording a USD debit, and apply the gross, net, short, asset and borrowing limits. Charge commissions, regulatory fees, slippage, margin interest and short-position costs.

Long-only and L/S returns are reported separately.

6 Walk-forward research design

MONTHLY WALK-FORWARD

At each month-end, calculate the signals and holdings with data dated on or before that decision. Apply those holdings only to the following month's returns, then record trading and financing costs.

CUT-OFF TESTS

Rerun the model through earlier cut-off dates. Signals, holdings and returns through each cut-off must match the same dates in the full run. Any mismatch fails the test.

RESEARCH WINDOWS

Portfolio rules and overlays were compared using returns through December 2018. The selected combination was then evaluated from January 2019 through May 2026. Both samples contain simulated monthly returns.

POINT-IN-TIME RERUN

The macro history was rebuilt with decision-date vintages for CPI, industrial production, payrolls and M2. The rerun through July 2026 and the L/S execution history remain retrospective simulations.

7 Risk limits and trading costs

Long-only limits

ETF and BIL weights total 100%. Exposure can fall as low as 50% when volatility is high; the remainder goes to BIL. Exposure cannot exceed 100%. Shorts and margin borrowing are disabled.

L/S limits

Maximum gross exposure: 175%. Maximum net exposure: 150%. Maximum gross shorts: 35%. Maximum absolute weight per non-cash ETF: 25%. Maximum USD margin debit: 50%. IBKR calculates the final Portfolio Margin requirement on the live account.

Execution cost assumptions

Assumptions reviewed August 20, 2026: \$0.005 per share, a \$1 order minimum, US regulatory-fee rates, 1 basis point of slippage and the applicable margin and short-proceeds tiers. It assumes 1% annual borrow for every short ETF because historical borrow data is unavailable. Taxes and market impact are excluded.